

Ledger: Making False Discovery Structurally Difficult in Autonomous Data Analysis

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Abstract

Language-model agents that analyse tabular data inherit a defect that predates them: they search a large space of relationships and report the ones that look interesting, without accounting for the size of the search. Given a table of pure noise, where the correct number of findings is exactly zero, such a system will confidently report several.

We present Ledger, a multi-agent analyst built around one architectural commitment: the language model proposes, and deterministic statistics decides. Candidate hypotheses are registered and hashed before any test executes, so the family over which error is controlled is fixed before any result is seen; test selection follows from checked assumptions rather than from model discretion; Benjamini–Hochberg correction is applied once across the entire registered family; and the natural-language report can only emit sentences the statistical layer has licensed.

We evaluate on 535 synthetic tables with ground truth known by construction, and report two results—one confirming the design, one undermining it.

First, session-level correction works. On 200 null tables an exhaustive uncorrected analyst—the behaviour of current agent pipelines—claims a false finding on 81.0% of tables; the same test machinery under one session-level correction claims one on 2.5%. The cost is six points of power and is concentrated entirely at small effects: above $d = 0.5$ it is under one percentage point.

Second, and less comfortably, pre-registration by a language model carries a severe hidden cost. When the registry is supplied by the proposer rather than by exhaustive enumeration, false discovery control becomes near-perfect (zero false findings across 25 null tables) but power collapses to 0.13. We localise the failure precisely: power conditional on a relationship having been registered is 0.92, while the proposer registers only 14% of the relationships that exist. The bottleneck is not the statistics but the blindness that makes the statistics valid—the proposer is restricted to marginal summaries precisely so that it cannot peek, and that restriction is what prevents it from finding anything. We quantify this tension, which we believe is intrinsic to pre-registration in autonomous agents, and release the generators, the

harness and the system.

1 Introduction

Exploratory data analysis is repetitive, skilled and slow, which makes it an obvious target for automation. Language models that write and execute code appear to make that automation trivial: point the system at a spreadsheet and let it report what it finds.

The appearance is misleading, for a reason older than language models. If a relationship is declared significant at $p < 0.05$, then by construction one in twenty tests on structureless data will pass. A table with 30 columns admits 435 pairwise relationships; roughly 22 of them will clear that threshold on pure noise. An agent that reports what it finds will therefore always find something, and will describe it fluently.

Human analysts are taught to guard against this with correction procedures, pre-registration and effect-size reporting. Automated systems have inherited the search behaviour without the discipline, and they operate at a scale that makes the problem worse: a tireless agent can test thousands of hypotheses in the time a human tests ten. Recent work confirms the concern is not hypothetical. Bertran et al. [2] show that autonomous LLM analysts reproduce the dispersion in effect sizes and conclusions observed in human many-analyst studies, and that the dispersion is steerable by changing the model or the assigned persona.

The premise of this work. Statistical discipline is not a feature to be added to an automated analyst afterwards. It has to be the thing that decides what the analyst is allowed to say. A system that can be talked out of its conclusions by a fluent narrative is not an analyst.

Contributions.

1. An architecture in which post-hoc selection is a run-time error, not a discouraged practice. Hypotheses are registered and hashed before any test executes; the registry is immutable thereafter. The family over which error is controlled is therefore fixed before any result is visible (§3).
2. Session-level false discovery control inside the agent. Benjamini–Hochberg is applied once across every

hypothesis registered in the session, not per test and not per insight type.

3. An evaluation protocol that measures whether an automated analyst correctly reports nothing. NULLSET and PLANTED are generated suites with ground truth known by construction rather than by adjudication (§4).
4. A measured decomposition of where the control comes from, a power curve stating what correction costs in sensitivity, and—the result we did not expect—a quantification of the coverage–validity tension that makes LLM pre-registration far more expensive than correction alone (§5).

We are explicit about what is not new here. Benjamini–Hochberg is from 1995 [1]; pre-registration as a methodological norm is older still. The contribution is the systems engineering that makes those ideas binding on an autonomous agent, together with the evaluation regime that shows it works.

2 Related Work

Automated analysis and insight discovery. AutoML systems such as Auto-sklearn [6] automate model selection but address prediction rather than explanation. Profiling tools of the ydata-profiling lineage generate exhaustive descriptive statistics with no notion of which findings matter; because they make no significance claims, they have no false-discovery exposure and no way to direct attention. QuickInsights [4] introduced automatic discovery of salient patterns and explicitly acknowledged the significance-testing problem, applying corrections within insight types—the closest pre-LLM work to our concern, though it cannot generate or execute novel analysis code.

Language-model data agents. InsightPilot [10] couples a model to an insight engine to steer exploration; Data Interpreter [9] demonstrates end-to-end agentic data-science pipelines with code execution. Both substantially advance what such systems can attempt. Neither controls false discovery across the exploration session, and neither prevents post-hoc selection of hypotheses. BLADE [8] benchmarks agents against expert-validated analytic decisions and documents that multiple justifiable approaches typically exist for a single question—a finding we return to in §6, because it bounds what any deterministic adjudicator can claim.

Per-test rigour versus session-level validity. The closest contemporary work is Fisher-R1 [11], which trains an agent by reinforcement learning for reliable hypothesis testing and introduces P-Bench, 425 tasks in which an agent must select a method, compute a p -value and

draw a conclusion. Fisher-R1 targets a failure mode adjacent to ours—inferential errors despite correctly executed analyses—and improves substantially on strong baselines. It is, however, scoped to a single hypothesis test per task, and neither its method nor its benchmark addresses multiple comparisons across a session. The distinction matters: per-test rigour is necessary but not sufficient. An agent that is individually correct on each of forty tests, with no accounting for the fact that it ran forty, still accumulates false discoveries at the rate our null experiments measure.

Detecting noise-fitting by other means. Rewolinski et al. [12] propose perturbation-based sanity checks grounded in the Predictability–Computability–Stability framework, which screen whether an agent can distinguish signal from noise and can expose affirmative conclusions as unsupported. This is a complementary axis: theirs is a stability argument applied after an analysis is produced, ours is a multiplicity argument enforced before any test runs. The two compose, and we regard combining them as the natural next step.

Statistical discipline and adaptivity. Benjamini–Hochberg [1] supplies the correction we apply. False-Positive Psychology [13] demonstrated how ordinary analytic flexibility produces significant results from nothing; the garden-of-forking-paths argument [7] showed this occurs even absent intent to game the analysis—precisely the situation of an autonomous agent. Theoretical work on adaptive data analysis [5] bounds generalisation error under adaptively chosen queries using differential privacy and holdout reuse. Our approach differs in kind: rather than tolerating adaptivity and paying for it with noise, we eliminate adaptivity by construction over the registered family. The resulting guarantee is narrower but legible and cheap to enforce.

Execution-grounded generation. Self-Debug [3] established that feeding execution results back to a model improves generated-code correctness. Ledger extends the principle from code correctness to claim correctness: execution decides not merely whether the code ran, but whether the sentence may be written.

3 The Ledger Architecture

Ledger is a fixed, hand-written state machine over eight agents. The orchestration is deliberately not a general-purpose agent framework: the ordering constraint between registration and testing is a correctness property, not a convenience, and must be inspectable. Table 1 lists the stages and, critically, which of them involve a language model at all.

Stage	Model?	Role
A0 Janitor	no	Type coercion, deduplication
A1 Profiler	no	Marginal summaries only
A2 Proposer	yes	Proposes testable hypotheses
A3 Registrar	no	Freezes and hashes the registry
A4 Executor	yes	Writes and repairs pandas code
A5 Statistician	no	Assumptions, test choice, FDR
A6 Reporter	yes	Writes prose from licensed text
A7 Adversary	yes	Red-teams the draft report

Table 1: The stages that decide what is true contain no language model. The model proposes and phrases; it never adjudicates.

3.1 The ordering constraint

A3 runs to completion before A4 starts. Once the registry is frozen no hypothesis may be added, and every registered hypothesis must be tested and reported on—including those that fail. This is what makes selective reporting impossible: the denominator of the correction is fixed before any result is seen. An agent that could append hypotheses mid-session, or quietly drop the disappointing ones, would invalidate the correction entirely.

The constraint is enforced at the data-model level rather than by prompt. The registry is a field on an immutable-after-freeze object, and insertion after freeze raises:

```
def add_hypothesis(self, entry):
    if self.is_frozen:
        raise ValueError(
            "Ledger is frozen. No hypotheses can be "
            "added after A3 completes.")
    self.hypotheses.append(entry)
```

A SHA-256 hash over the registered statements is computed at the freeze and reported with the analysis, so a reader can verify that the family which was corrected is the family that was registered.

3.2 Deterministic adjudication

A5 contains no language model. Given a hypothesis and a data slice it selects the test from checked properties rather than from a model’s suggestion: Shapiro–Wilk and Levene decide between Welch’s t and Mann–Whitney U ; expected cell counts decide between χ^2 and Fisher’s exact test; for $k > 2$ groups, normality and homogeneity decide between one-way ANOVA and Kruskal–Wallis. Every result carries an effect size—Cohen’s d , Cramér’s V , η^2 —because significance without magnitude is not a finding.

After all registered tests complete, Benjamini–

Hochberg is applied once across the entire family at $q = 0.05$. Survivors become claims; everything else is recorded as tested-and-not-supported and appears in an appendix rather than vanishing.

3.3 Claim-level provenance

Each adjudicated hypothesis yields a `licensed_text` field, the only text A6 may use for that hypothesis. It cannot strengthen the wording, add a causal reading, or mention a hypothesis with no entry. In the interface, every sentence of the report links back to the ledger entry that licensed it, so a reader can move from any claim to the code that produced it, the assumptions that were checked, and the test those assumptions selected.

4 Evaluation Protocol

The clearest way to evaluate an automated analyst is to give it a dataset with nothing in it and count what it claims to find. The correct number is zero, and it is zero by construction rather than by judgement.

NULLSET. 200 tables, 300–2500 rows and 6–14 columns, drawn independently. Columns carry plausible names, mixed dtypes, skewed and heavy-tailed numerics (log-normal, gamma, bimodal, count), unbalanced categoricals and missingness up to 12%, so that the tables are not distinguishable from real ones by inspection. A separate 40-table variant adds one deterministic nuisance coupling—a derived column—which is genuinely real but analytically uninteresting; it is recorded as ground truth so a system is not penalised for finding it.

PLANTED. 240 tables containing 2–4 genuine relationships at controlled effect sizes $d \in \{0.2, 0.35, 0.5, 0.8\}$, buried among noise columns. Effects are planted in three forms so that test selection is exercised rather than a single code path: a mean shift on a binary split, a linear association between two numerics (with $r = d/\sqrt{d^2 + 4}$), and a spread of group means across a $k > 2$ factor.

Arms. All arms execute the same test-selection code—A5’s own selector. This is deliberate and is what makes the comparison meaningful: the arms differ only in which hypotheses enter the family and whether the family is corrected, never in how a test is chosen or computed.

- Exhaustive, uncorrected. Test every column pair; claim every $p < 0.05$. This is what an agent with code execution and no discipline does.
- Exhaustive, BH-corrected. Every column pair, corrected once across all of them. Isolates the contri-

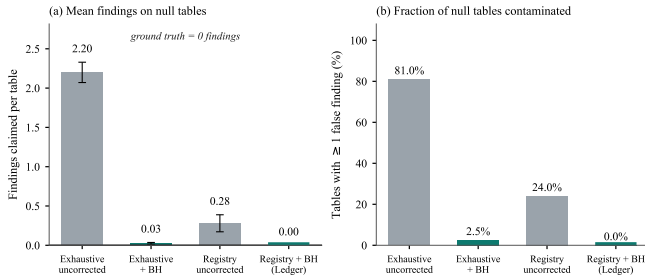


Figure 1: Findings claimed on 200 tables containing no real relationships. Ground truth is zero. Error bars are standard errors over tables.

bution of correction alone, holding the candidate set fixed.

- Registry, uncorrected and Registry, BH-corrected. The candidate set is whatever the A2 proposer registered from the marginal profile, frozen before execution. Comparing these against the exhaustive arms separates the effect of a smaller, model-chosen family from the effect of correcting across it.

Failed tests remain in the family at $p = 1.0$ rather than being dropped; removing them would shrink m and make the threshold more permissive for everything else, which is the exact accounting error the architecture exists to prevent.

Measures. We report mean findings claimed per table; the fraction of tables on which at least one false finding is claimed (the quantity a practitioner cares about); the false discovery proportion $FDP = FP / \max(\text{claims}, 1)$, whose expectation BH controls; and, on PLANTED, power as recall of planted effects.

5 Results

Across the suites, 23,901 statistical tests were executed over 23,909 candidate hypotheses.

5.1 The null-dataset result

Figure 1 and Table 2 give the headline. On tables where the correct number of findings is zero, the uncorrected exhaustive analyst claims 2.20 ± 0.25 findings per table and contaminates 81.0% of tables with at least one false discovery. Applying Benjamini–Hochberg once across the same candidate set, using the same tests on the same data, reduces this to 0.025 ± 0.022 claims per table and 2.5% of tables—a 32 $\times$ reduction in claims and a 98% reduction in the fraction of contaminated analyses.

We draw attention to the nuisance variant. When a single trivially-real relationship is present, the corrected system recovers it on every table (power 1.00) while claiming 1.08 things in total; the uncorrected system recovers it too but buries it among 3.03 false find-

Suite	Arm	Claims	Any FD	FDP
NULLSET	uncorrected	2.200	81.0%	0.810
	BH	0.025	2.5%	0.025
NULLSET + nuisance	uncorrected	4.025	95.0%	0.668
	BH	1.075	5.0%	0.029
PLANTED	uncorrected	5.287	85.8%	0.410
	BH	2.850	19.6%	0.056

Table 2: Mean claims per table, fraction of tables with at least one false discovery, and mean false discovery proportion. $n = 200/40/240$ tables.

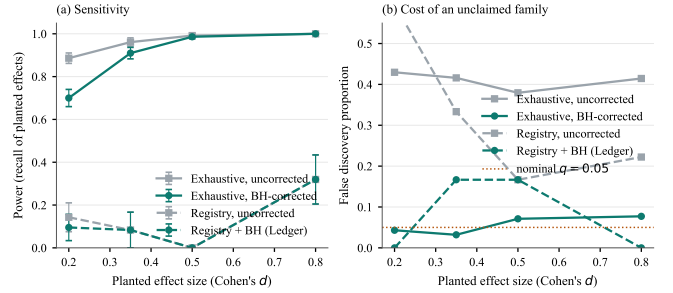


Figure 2: Power and false discovery proportion against planted effect size on 240 PLANTED tables. The corrected arm tracks the nominal $q = 0.05$ line; the uncorrected arm sits near 0.41 throughout.

ings. Correction does not merely suppress output—it suppresses the wrong output.

5.2 What discipline costs

A system that reports nothing is trivially safe and useless, so the interesting question is what the discipline costs in sensitivity. Figure 2 answers it. Aggregated over PLANTED, correction reduces power from 0.960 to 0.899—six percentage points—while reducing the false discovery proportion from 0.410 to 0.056.

The aggregate understates how favourable the trade is, because the cost is concentrated entirely at small effects:

d	Power (BH)	Power (unc.)	Δ
0.20	0.700	0.886	−0.186
0.35	0.910	0.961	−0.051
0.50	0.986	0.992	−0.006
0.80	1.000	1.000	0.000

At $d \geq 0.5$ correction is effectively free: it costs under one percentage point of power and takes the false discovery proportion from ~ 0.40 to ~ 0.07 . The system pays for its discipline only where the signal is genuinely faint, which is where a finding is least trustworthy anyway. We regard this as the central practical result: the objection that statistical discipline makes an automated analyst too conservative to be useful is, at moderate and large effect sizes, simply not borne out.

Family	Corr.	m	Any FD	Power
Exhaustive	no	51.6	85.8%	0.960
Exhaustive	BH	51.6	19.6%	0.899
Registry	no	8.7	40.0%	0.137
Registry	BH	8.7	8.0%	0.123

Table 3: PLANTED. A model-chosen registry drives false discovery down further than correction alone, and takes power with it.

5.3 Empirical calibration

The corrected arm’s false discovery proportion is 0.043, 0.032, 0.071 and 0.077 at $d = 0.2, 0.35, 0.5, 0.8$ against a nominal $q = 0.05$. That the achieved rate sits near the nominal rate, rather than far below it, indicates the procedure is neither broken nor gratuitously conservative on these suites.

5.4 The cost of pre-registration

The arms above hold the candidate family fixed at every column pair. That isolates correction, but it is not how Ledger runs: in the deployed system the family is whatever the proposer registered. Substituting the model-chosen registry produces the result in Table 3, and it is not the result we expected.

On null tables the registry arm is flawless: across 25 tables it claimed zero false findings, against 2.5% of tables for exhaustive-plus-BH. A smaller family is a genuinely powerful error-control mechanism, and combining it with correction is better than either alone.

But power falls from 0.899 to 0.123. A system that finds one planted relationship in eight is not an analyst. The obvious reading—that discipline has made the system too conservative—turns out to be wrong, and the instrumentation says why. Over 30 further PLANTED tables containing 92 planted relationships in total:

Planted relationships	92
... the proposer registered	13 (14.1%)
... the system then detected	12
Unconditional power	0.130
Power conditional on registration	0.923

Once a real relationship is in the registry, the pipeline finds it 92% of the time. Nothing is wrong with the tests, the assumption checks, or the correction. The entire loss is upstream: the proposer registers about nine of roughly fifty-two available pairs, and those nine are mostly not the ones that carry signal.

Why this is not a prompt-engineering bug. It would be natural to conclude that the proposer simply needs a better prompt. We think the problem is structural. The proposer is shown marginal summaries only—types, cardinality, missingness, per-column distributions—and that restriction is not incidental. It is what makes the

correction sound: if hypothesis selection were informed by the relationships in the data, the registry would be a data-dependent choice and the family would no longer be fixed independently of the results, exactly the leakage that FDR control assumes away. So the proposer is blind by construction, and a blind proposer choosing nine pairs out of fifty-two will cover about seventeen percent of the space regardless of how well it is prompted.

This is the coverage–validity tension, and we state it plainly because it constrains the whole design: the restriction that makes pre-registration statistically meaningful is the same restriction that prevents the proposer from knowing where to look. Any system that pre-registers hypotheses through a model that has not seen the relationships will face some version of it.

What this implies. The tension is a budgeting problem, not a refutation. Three directions follow, in increasing order of interest. The registration budget can simply be raised—registering thirty pairs instead of nine trades a more conservative threshold for much better coverage, and because BH scales gracefully in m this trade is cheap. Domain knowledge, supplied through the data dictionary rather than through the data, raises coverage without creating leakage, which is an argument for the RAG path the system already has. Most interestingly, sample splitting would permit a proposer to see relationships in one partition and register hypotheses tested on another, recovering coverage while keeping the family independent of the data it is evaluated on; this is the adaptive-data-analysis setting [5] and connecting the two properly is the clearest next step for this work.

6 Threats to Validity

Synthetic nulls may be easier than real ones. Real data contains genuine but uninteresting structure—encoding artefacts, collection effects—that independently drawn columns lack. Our nuisance variant is a partial mitigation and shows the corrected system handles one such coupling correctly, but a stronger test is permuted real datasets, which preserve marginal distributions while destroying relationships. We have not yet run that and do not claim its result.

FDR control assumes the registry is the full family. If the proposer’s choice of hypotheses is itself informed by peeking at the data, the correction is optimistic. The profiler’s output is deliberately restricted to marginal summaries for this reason, and the restriction is enforced in code rather than requested in a prompt. We have not measured the residual leakage, and we flag it as the most important open question about the guarantee.

The main experiment isolates correction, not the full architecture. The two arms reported here share a can-

didate set and differ only in correction. This cleanly establishes what correction contributes but does not, on its own, establish what pre-registration contributes over and above it. A proposer-driven arm is required for that, and is discussed in §7.

Deterministic does not mean uniquely correct. BLADE [8] documents that experts frequently disagree about the right analysis for a given question. Our claim for A5 is therefore that test choice is reproducible and inspectable, not that it is the only defensible choice. Where assumptions do not resolve cleanly, the selected test and the checks that selected it are both recorded.

Results are implementation-dependent. All arms share one test implementation. This is a strength for internal validity and a limit on external validity: a different statistics stack might shift absolute numbers, though the direction of the comparison does not depend on it.

7 Limitations and Future Work

The registry arms are run at $n = 25$ null and $n = 55$ planted tables against $n = 480$ for the deterministic arms, because each costs a model call. The coverage figure of §5.4 is therefore the least precisely estimated number in the paper, and the one most worth re-running at scale. Its direction is not in doubt—a proposer that registers nine of fifty-two pairs cannot have high coverage—but the point estimate should be read with that sample size in mind.

We evaluate one proposer model. Coverage is plausibly model-dependent in a way that correction is not, so the 14% figure should be read as a measurement of this configuration rather than a constant of nature.

Four extensions follow directly. First, the registration-budget sweep of §5.4, which is cheap and would turn the tension we report into a dose-response curve. Second, sample splitting to break the tension outright, connecting this work to adaptive data analysis [5]. Third, evaluation on an external benchmark: P-Bench [11] is the natural target, extended so that multiple tasks are bundled into one session, which would test session-level validity on data we did not generate. Fourth, composing our multiplicity argument with the stability argument of Rewolinski et al. [12]; the two screen for different failures and neither subsumes the other.

8 Conclusion

Automated analysts have inherited the search behaviour of human exploratory analysis without its discipline. We have shown that the discipline can be made architectural rather than advisory, and that doing so takes the fraction

of null tables yielding a false finding from 81.0% to 2.5% at a cost in power that is negligible above $d = 0.5$.

We have also shown that the two mechanisms are not equally cheap. Correction is close to free; pre-registration through a blind proposer is not, and costs roughly seven-eighths of the system’s power in our configuration—not because the statistics are too strict but because the proposer covers 14% of the search space. We would rather report that than not. It is the more useful finding of the two: it says where the next improvement has to come from, and a version of this paper that omitted it would have described a system that controls false discovery beautifully and finds almost nothing.

The system, the generators and the harness are available at https://github.com/garvbahl37-gif/Ledger_agent.

References

- [1] Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. Royal Statistical Society B*, 57(1):289–300, 1995.
- [2] M. Bertran, R. Fogliato, and Z. S. Wu. Many AI analysts, one dataset: navigating the agentic data science multiverse. *arXiv:2602.18710*, 2026.
- [3] X. Chen, M. Lin, N. Schärli, and D. Zhou. Teaching large language models to self-debug. In *ICLR*, 2024.
- [4] R. Ding, S. Han, Y. Xu, H. Zhang, and D. Zhang. QuickInsights: quick and automatic discovery of insights from multi-dimensional data. In *ACM SIGMOD*, 2019.
- [5] C. Dwork, V. Feldman, M. Hardt, T. Pitassi, O. Reingold, and A. Roth. Preserving statistical validity in adaptive data analysis. In *ACM STOC*, 2015.
- [6] M. Feurer, A. Klein, K. Eggenberger, J. Springenberg, M. Blum, and F. Hutter. Efficient and robust automated machine learning. In *NeurIPS*, 2015.
- [7] A. Gelman and E. Loken. The statistical crisis in science. *American Scientist*, 102(6):460–465, 2014.
- [8] K. Gu, R. Shang, R. Jiang, K. Kuang, R.-J. Lin, D. Lyu, Y. Mao, Y. Pan, T. Wu, J. Yu, Y. Zhang, T. M. Zhang, L. Zhu, M. A. Merrill, J. Heer, and T. Althoff. BLADE: benchmarking language model agents for data-driven science. In *EMNLP*, 2024. *arXiv:2408.09667*.
- [9] S. Hong et al. Data interpreter: an LLM agent for data science. *arXiv:2402.18679*, 2024.
- [10] P. Ma, R. Ding, S. Han, and D. Zhang. InsightPilot: an LLM-empowered automated data exploration system. In *EMNLP (System Demonstrations)*, 2023.
- [11] J. Miao, J. Mu, G. Chen, and J. Zou. Fisher-R1: training LLM agents for reliable hypothesis testing. *arXiv:2608.07437*, 2026.
- [12] Z. T. Rewolinski, A. V. Zane, H. Huang, C. Singh, C. Wang, J. Gao, and B. Yu. Sanity checks for agentic data science. *arXiv:2604.11003*, 2026.

- [13] J. Simmons, L. Nelson, and U. Simonsohn. False-positive psychology: undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011.